

Zonotope faces are not counted by arrangement regions

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10 October 2026

Disproof of conjecture 00000007132. The conjecture equates the face count of a zonotope with the region count of its hyperplane arrangement. The square already refutes this for total faces and even for nonempty proper faces.

Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$ in $\mathbb{R}^2$. The zonotope

$$Z = [-e_1, e_1] + [-e_2, e_2] = [-1, 1]^2$$

is the Minkowski sum of two actual line segments: each point (x, y) is the sum $(x, 0) + (0, y)$, and these sums are precisely the points satisfying $-1 \leq x, y \leq 1$.

The associated central hyperplane arrangement consists of $e_1^\perp = \{x = 0\}$ and $e_2^\perp = \{y = 0\}$. Its complement has four regions, namely the four open quadrants. Each quadrant is convex and nonempty, hence connected. A connected subset of the complement cannot contain points with opposite signs in either coordinate: a coordinate projection is continuous, and the intermediate value theorem would give a point with coordinate zero. Consequently the four quadrants are exactly the connected components, and the region count is 4.

The nonempty proper faces of Z are its four vertices and its four edges, for a total of 8. These are genuine exposed faces: a face is the set of maximizers of a linear functional $ax + by$ on Z . If both coefficients are nonzero, the unique maximizing vertex is $(\operatorname{sgn} a, \operatorname{sgn} b)$. If exactly one coefficient is zero, the maximizers form the corresponding edge. If both are zero, the face is Z itself. This also classifies all exposed faces of this polytope. The usual total nonempty face count is 9; counting the empty face gives 10. None of 8, 9, 10 equals 4.

For the contradiction it suffices to exhibit five distinct nonempty proper faces: the right, left, top, and bottom edges, together with $(1, 1)$. Their exposing coefficient pairs are respectively

$$(1, 0), \quad (-1, 0), \quad (0, 1), \quad (0, -1), \quad (1, 1).$$

All are different and are strict subsets of the square.

The valid related theorem. The familiar region correspondence counts *vertices* of the zonotope, rather than all its faces. Here both the vertex count and the arrangement region count are 4. We do not refute that theorem. The characteristic polynomial of the two-axis arrangement is $(q - 1)^2$, whose evaluation at $q = -1$ gives 4 regions; it does not give the total face count claimed in the source.

Lean coverage. The project defines the actual square and its generating coordinate segments and proves the Minkowski-sum equality. It uses the standard argmax definition of exposed faces of real linear functionals, proves five explicit exposing identities, injectivity of the five-face family, and properness and nonemptiness. It defines the four actual quadrant sets, proves each connected, and applies the intermediate value theorem to prohibit axis crossings by any connected subset of the complement. The accompanying explanation identifies these sets with the four regions. Lean 4.19.0 and Mathlib are pinned, and printed audits use only standard logical axioms.